

Baseline, Parametric, and Operator Learning Models for One-Dimensional Advection–Dispersion

Ali Haghighi

Supervised by:

Afshin Ashrafzadeh, François Lehmann, Marwan Fahs

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Note, Apr 12, 2026, written after sharing report. After this report was completed, we reran all three models with a regular collocation mesh (50 points per axis) and L-BFGS as the only optimizer. Training was substantially faster and produced near-perfect agreement with the reference in each case. We did not document that follow-up in a separate report, because attention had already shifted to new problems.

Abstract

This report re-establishes the one-dimensional advection–dispersion study under a revised problem setup designed to restore a meaningful comparison with the Ogata–Banks analytical solution by keeping the plume sufficiently far from the outlet over the simulated time horizon. Within this updated setting, three physics-informed surrogates are documented: a baseline PINN for a fixed transport regime, a parametric PINN over $Pe \in [50, 5000]$, and a parametric DeepONet-style PINO over the same family. The report details the governing formulation, the shift from advective to physical time scaling, the training losses, and the collocation and tensor structures used for each model. All three selected runs achieve near-perfect agreement with the analytical reference, with the PINO giving the closest match among the reported cases. At the same time, because the models were obtained under different training strategies, the present work should be read as a re-establishment of accurate models rather than a controlled comparison of methods. The next steps is therefore to turn these exploratory findings into a more systematic and reproducible training methodology.

1 Introduction

Since the last meeting, the baseline PINN and parametric PINN have both reached a robust level of accuracy, and exploratory work on a parametric operator model (PINO) has also advanced substantially. In the process, an important issue became clear: under the earlier problem settings, the finite-domain solution could remain nonzero at the outlet over the simulated time horizon, while the Ogata–Banks solution assumes a semi-infinite setting in which the far boundary does not influence the solution. This created a mismatch between the model outcome and the analytical reference. To address this, the transport parameters were revised so that the plume remains sufficiently far from the outlet, allowing Ogata–Banks to again serve as a meaningful benchmark. Under this revised setup, the present report re-establishes the study from the ground up, beginning with the baseline PINN, then the parametric PINN, and finally a parametric PINO.

I have also explored a wide range of training choices across the three models, with accuracy as the main priority. That objective has now been achieved for the selected runs reported here. However, these experiments were exploratory rather than fully systematic: they have already yielded useful practical insight, but they do not yet provide a rigorous basis for concluding which training strategy is best overall. That question remains the focus of the ongoing work. From the many exploratory experiments conducted so far, the clearest findings

are:

- Collocation design is central to performance. The total number of collocation points, their sampling distribution, and their resampling frequency during training all strongly affect solution quality.
- Uniformly random collocation with periodic resampling has consistently performed better than regular-grid sampling.
- Latin hypercube sampling did not provide a clear advantage over uniformly random sampling.
- Small collocation budgets combined with repeated random resampling can still produce overfitted or unstable final solutions.
- For the PINO formulation, increasing the sensor count did not improve accuracy and only introduced unnecessary architectural complexity; for the present problem, a single sensor corresponding to $\log Pe$ is sufficient.

This report presents three selected runs that achieve near-perfect results under the revised problem setup. Section 2 defines the problem, normalization, loss formulation, and model-specific training procedures. Section 3 then presents the baseline PINN, parametric PINN, and parametric PINO results, followed by next steps and supporting appendices.

2 Methods

This section presents the governing equation and boundary conditions, the normalization and time scaling used during training, the composite physics-informed loss, and the three model classes considered here: a single-regime baseline PINN, a parametric PINN over dispersion (equivalently Péclet number), and a parametric DeepONet-style PINO over the same family. A compact summary of the main training settings—inputs, architectures, optimizers, and collocation budgets—is given at the end of the section in Table 2.

2.1 Problem setup

We consider one-dimensional solute transport governed by the physical advection–dispersion equation

$$\frac{\partial C}{\partial t} + U \frac{\partial C}{\partial x} = D \frac{\partial^2 C}{\partial x^2}, \quad (x, t) \in [0, L] \times [0, T_{\text{phys}}], \quad (1)$$

with initial and boundary conditions

$$C(x, 0) = 0, \quad C(0, t) = C_0, \quad C(L, t) = 0. \quad (2)$$

The physical parameters used in the reported runs are listed in Table 1. The baseline PINN is trained for a single dispersion coefficient, whereas the parametric PINN and PINO are trained over a dispersion interval and therefore over a corresponding range of Péclet numbers,

$$Pe = \frac{UL}{D}. \quad (3)$$

The analytical Ogata–Banks solution was used only for post-training visual comparison and never during optimization [1]. Under the present parameter choices, the plume remains suf-

ficiently far from the outlet over the simulated time horizon, making the Ogata–Banks profile an appropriate benchmark for the reported comparisons.

Table 1: Physical parameters used in the reported runs.

Symbol	Value
U	0.05 m day ⁻¹
L	100 m
T_{phys}	1000 day
C_0	5 kg m ⁻³
D	0.001 to 0.1 m ² day ⁻¹ *
$\text{Pe} = UL/D$	50 to 5000*

*For the baseline PINN only, $D = 0.1 \text{ m}^2 \text{ day}^{-1}$, corresponding to $\text{Pe} = 50$.

2.2 Normalization and time scaling

Earlier PINN formulations in this project, as well as the related technical reports, used *advective* time scaling with $T_s = L/U$. In the present study, and in ongoing work, we instead use *physical* time scaling, in which the reference time is the simulated horizon T_{phys} . All three models reported here—the baseline PINN, parametric PINN, and parametric PINO—use this convention.

To improve conditioning, all models are trained on normalized variables,

$$x^* = \frac{x}{L}, \quad t_s = \frac{t}{T_s}, \quad C^* = \frac{C}{C_0}, \quad (4)$$

where T_s is the chosen time scale. Substituting these definitions into the physical equation yields the general scaled form

$$\frac{\partial C^*}{\partial t_s} + \frac{UT_s}{L} \frac{\partial C^*}{\partial x^*} = \frac{DT_s}{L^2} \frac{\partial^2 C^*}{\partial x^{*2}}. \quad (5)$$

Two choices of T_s are relevant.

Advective time scaling. With the classical advective scale

$$T_{\text{adv}} = \frac{L}{U}, \quad (6)$$

Eq. (5) becomes

$$\frac{\partial C^*}{\partial t_{\text{adv}}} + \frac{\partial C^*}{\partial x^*} = \frac{1}{\text{Pe}} \frac{\partial^2 C^*}{\partial x^{*2}}. \quad (7)$$

For the present parameters, $L = 100 \text{ m}$ and $U = 0.05 \text{ m day}^{-1}$, giving

$$T_{\text{adv}} = \frac{100}{0.05} = 2000 \text{ day}. \quad (8)$$

Since the simulated horizon is $T_{\text{phys}} = 1000 \text{ day}$, the normalized time presented to the network would span only

$$t_{\text{adv}}^* \in \left[0, \frac{T_{\text{phys}}}{T_{\text{adv}}}\right] = [0, 0.5]. \quad (9)$$

Physical time scaling. Here we instead take

$$T_s = T_{\text{phys}} = 1000 \text{ day}, \quad (10)$$

so that

$$t^* = \frac{t}{T_{\text{phys}}} \in [0, 1]. \quad (11)$$

Equation (5) then becomes

$$\frac{\partial C^*}{\partial t^*} + \frac{UT_{\text{phys}}}{L} \frac{\partial C^*}{\partial x^*} = \frac{DT_{\text{phys}}}{L^2} \frac{\partial^2 C^*}{\partial x^{*2}}. \quad (12)$$

For the present parameter set,

$$\frac{UT_{\text{phys}}}{L} = \frac{0.05 \times 1000}{100} = 0.5, \quad (13)$$

so the residual enforced during training is

$$\frac{\partial C^*}{\partial t^*} + 0.5 \frac{\partial C^*}{\partial x^*} = \frac{0.5}{\text{Pe}} \frac{\partial^2 C^*}{\partial x^{*2}}. \quad (14)$$

This change does *not* alter the physical problem; it only changes the normalized coordinates and the coefficients appearing in the scaled residual. Its practical advantage is that the network sees the full temporal input range $t^* \in [0, 1]$, rather than the compressed interval $[0, 0.5]$ produced by advective scaling. In practice, this gives a better-conditioned input space and improves training, particularly at early times.

2.3 Shared optimization and loss

All three models are trained by minimizing the sum of a PDE residual loss, an initial-condition loss, an inlet boundary loss, and an outlet boundary loss,

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{IC}} + \mathcal{L}_{\text{inlet}} + \mathcal{L}_{\text{outlet}}. \quad (15)$$

2.4 Baseline PINN

The baseline PINN was trained for a single transport regime,

$$\begin{aligned} U &= 0.05 \text{ m day}^{-1}, & D &= 0.1 \text{ m}^2 \text{ day}^{-1}, \\ L &= 100 \text{ m}, & C_0 &= 5 \text{ kg m}^{-3}, \end{aligned} \quad (16)$$

corresponding to $\text{Pe} = 50$.

The network is a fully connected multilayer perceptron with architecture

$$[2, 16, 16, 16, 16, 1],$$

that is, four hidden Tanh layers of width 16 mapping input (x^*, t^*) to output C^* . Training uses Adam only, with learning rate 10^{-3} for 15,000 epochs.

At each epoch, a fresh uniformly sampled collocation set is drawn. The interior PDE set contains $40^2 = 1600$ random points in (x^*, t^*) . The initial-condition set contains 40 random points at $t^* = 0$, the inlet set contains 40 random points at $x^* = 0$, and the outlet set contains 40 random points at $x^* = 1$.

2.5 Parametric PINN

The parametric PINN extends the baseline formulation to a family of transport problems with

$$D \in [0.001, 0.1] \text{ m}^2 \text{ day}^{-1}, \quad \text{Pe} \in [50, 5000]. \quad (17)$$

The model input is $(x^*, t^*, \log \text{Pe})$, and the output remains the scalar concentration C^* . Its architecture is

$$[3, 16, 16, 16, 16, 1],$$

that is, four hidden Tanh layers of width 16.

Training is performed in two stages. The first stage uses Adam for 14,000 epochs with learning rate 10^{-3} . The second stage uses L-BFGS with learning rate 1.0 for 1000 steps.

During Adam, collocation is resampled every epoch. During L-BFGS, a fresh draw is used whenever the loss is recomputed. Each interior PDE set contains $40^3 = 64000$ uniformly sampled random triplets $(x^*, t^*, \log \text{Pe})$, stored as a single tensor of shape $\mathbb{R}^{64000 \times 3}$. The IC, inlet, and outlet sets each contain 40 random samples, with tensor shape $\mathbb{R}^{40 \times 3}$.

2.6 Parametric PINO (DeepONet)

The third model is a parametric neural-operator formulation implemented as a DeepONet-style PINO [2] over the same parameter range,

$$D \in [0.001, 0.1] \text{ m}^2 \text{ day}^{-1}, \quad \text{Pe} \in [50, 5000]. \quad (18)$$

Its architecture separates parameter encoding from coordinate encoding. The branch network acts on the scalar parameter input and has widths

$$[1, 16, 16, 16],$$

while the trunk network acts on the space-time coordinates and has widths

$$[2, 16, 16, 16].$$

Their latent outputs are combined pointwise to produce the concentration prediction.

Training again follows a two-stage schedule. The Adam stage uses 5000 epochs with learning rate 10^{-3} . The L-BFGS refinement stage uses learning rate 1.0 for 5000 steps.

During Adam, collocation is resampled every epoch. Each draw contains $20^3 = 8000$ random triplets $(x^*, t^*, \log \text{Pe})$ for the interior PDE residual, implemented as a trunk tensor of shape $\mathbb{R}^{8000 \times 2}$ and a matching branch tensor of shape $\mathbb{R}^{8000 \times 1}$. The IC, inlet, and outlet sets are likewise resampled every epoch, with 20 random points each.

During L-BFGS, a *single* uniformly sampled collocation set is drawn once before refinement: $50^3 = 125000$ interior triplets, implemented as a trunk tensor $\mathbb{R}^{125000 \times 2}$ and a branch tensor $\mathbb{R}^{125000 \times 1}$, together with 50 samples for each of the IC, inlet, and outlet constraints. This set is held fixed throughout the full L-BFGS stage.

2.7 Collocation sets and tensor representation

For clarity, counts such as 40^2 , 40^3 , 20^3 , or 50^3 refer only to the *total number* of uniformly sampled random collocation points. They do *not* denote structured tensor-product grids.

Thus, $40^2 = 1600$ interior points in the baseline PINN means 1600 independently sampled random pairs (x^*, t^*) . Likewise, $40^3 = 64000$ in the parametric PINN means 64000 independently sampled random triplets $(x^*, t^*, \log \text{Pe})$, not a Cartesian $40 \times 40 \times 40$ lattice.

The tensor organization differs between the PINN and PINO implementations:

Baseline PINN. The interior PDE collocation set is stored as a single tensor

$$\mathbf{X}_{\text{PDE}} \in \mathbb{R}^{N \times 2}, \quad (19)$$

whose rows are $[x_i^*, t_i^*]$. For the reported run, $N = 40^2 = 1600$. The initial-condition and boundary-condition tensors each have shape $\mathbb{R}^{40 \times 2}$, with one coordinate fixed and the other sampled uniformly at random.

Parametric PINN. The interior PDE collocation set is again a single tensor, now with three columns,

$$\mathbf{X}_{\text{PDE}} \in \mathbb{R}^{N \times 3}, \quad (20)$$

whose rows are $[x_i^*, t_i^*, \log \text{Pe}_i]$. For the reported run, $N = 40^3 = 64000$. The IC, inlet, and outlet tensors each have shape $\mathbb{R}^{40 \times 3}$. In other words, the parametric PINN concatenates space, time, and parameter into one input tensor before the first layer.

PINO / DeepONet. The PINO does *not* concatenate $(x^*, t^*, \log \text{Pe})$ into one tensor. Instead, it takes two separate inputs,

$$\mathbf{Y}_{\text{PDE}} \in \mathbb{R}^{N \times 2}, \quad \mathbf{P}_{\text{PDE}} \in \mathbb{R}^{N \times 1}, \quad (21)$$

where the rows of $\mathbf{Y}_{\text{PDE}}$ are $[x_i^*, t_i^*]$ and the rows of $\mathbf{P}_{\text{PDE}}$ are $[\log \text{Pe}_i]$. The trunk network acts on $\mathbf{Y}_{\text{PDE}}$, the branch network acts on $\mathbf{P}_{\text{PDE}}$, and the two latent representations are combined only after these separate encodings. This is the key implementation difference between the parametric PINN and the PINO.

3 Results

The three models reported below were trained with different optimization and sampling strategies and were selected from a broader set of completed and ongoing experiments. All three achieve near-perfect agreement with the analytical reference, but the present results should not be read as basis for comparison between architectures.

3.1 Baseline PINN

Figure 1 shows the baseline PINN for the fixed case $D = 0.1$ ($\text{Pe} = 50$). The agreement with the Ogata–Banks reference is near perfect across all plotted times. The only visible imperfection is a very slight underprediction of the concentration profile.

3.2 Parametric PINN

Figure 2 shows the parametric PINN at $\text{Pe} = 50, 500$, and 5000. Overall, the model remains very accurate across the full parameter range.

At $\text{Pe} = 50$, the result is again near perfect, with a slight underprediction similar to the baseline case but marginally more pronounced. At $\text{Pe} = 500$, the profiles are also near

Table 2: Training summary. Exponents such as 40^2 or 50^3 denote the *total number* of i.i.d. uniformly sampled points, not a tensor-product grid. All networks use fully connected MLP blocks with Tanh hidden activations; the PINO uses separate DeepONet-style branch and trunk networks before the final readout. Collocation is resampled every Adam epoch. During L-BFGS, the parametric PINN redraws points whenever the loss is recomputed, whereas the PINO uses one fixed collocation draw for the entire L-BFGS stage.

Model	Inputs	Layer widths	Optimizer	Interior N	IC / inlet / outlet
Baseline PINN	$\mathbf{X} \in \mathbb{R}^{N \times 2}$, rows $[x^*, t^*]$	[2, 16, 16, 16, 16, 1]	Adam, 15,000 epochs, $lr = 10^{-3}$	$40^2 = 1600$, resampled each epoch	40 each, resampled each epoch
Parametric PINN	$\mathbf{X} \in \mathbb{R}^{N \times 3}$, rows $[x^*, t^*, \log Pe]$	[3, 16, 16, 16, 16, 1]	Adam, 14,000 epochs, $lr = 10^{-3}$; L-BFGS, 10^3 steps, $lr = 1$	$40^3 = 64,000$; resampled each Adam epoch and at each L-BFGS loss recomputation	40 each; same resampling policy as interior points
Parametric PINO	Trunk $\mathbf{Y} \in \mathbb{R}^{N \times 2}$, branch $\mathbf{P} \in \mathbb{R}^{N \times 1}$	Branch [1, 16, 16, 16]; trunk [2, 16, 16, 16]	Adam, 5000 epochs, $lr = 10^{-3}$; L-BFGS, 5000 steps, $lr = 1$	Adam: $20^3 = 8000$, resampled each epoch; L-BFGS: $50^3 = 125,000$, sampled once	Adam: 20 each, resampled each epoch; L-BFGS: 50 each, sampled once

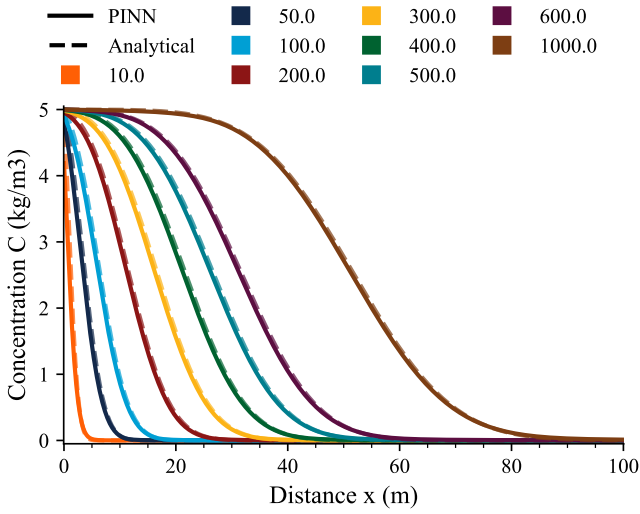


Figure 1: Baseline PINN solution with fixed dispersion $D = 0.1$, which corresponds to $Pe = 50$ under the reference velocity and length used in training.

perfect; the only visible imperfection is a very small deviation at the beginning of the front, where the concentration starts to drop. At $Pe = 5000$, this same feature remains very small, and the trailing part of the front, where the concentration flattens toward zero, is also slightly overpredicted.

3.3 Parametric PINO

Figure 3 presents the corresponding PINO/DeepONet predictions at the same three Péclet numbers. Among the three reported models, this one is the most accurate, with agreement that is even slightly closer to the analytical reference than in the baseline and parametric PINN cases.

4 Conclusion and Next Steps

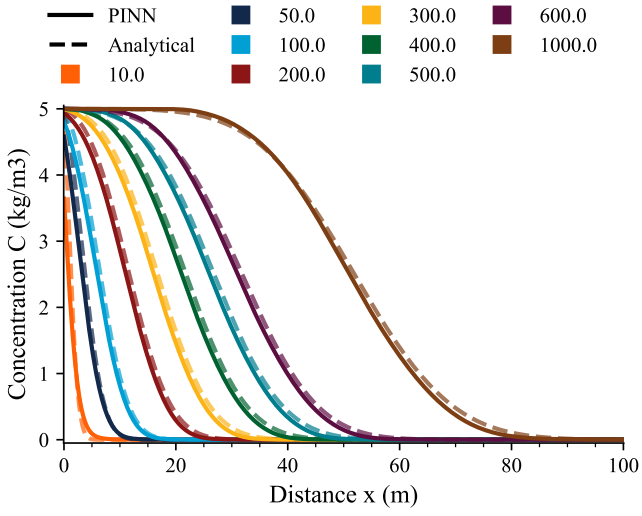
Under the revised problem setup, all three selected models—the baseline PINN, parametric PINN, and parametric PINO—achieve near-perfect agreement with the Ogata–Banks reference. This confirms that the updated parameter regime and training formulations are now sufficient to recover highly accurate solutions across both the fixed and parametric settings.

At the same time, the reported runs were obtained under different optimization and sampling strategies, so the present report should be read as a re-establishment of accurate models rather than as a controlled comparison of methods.

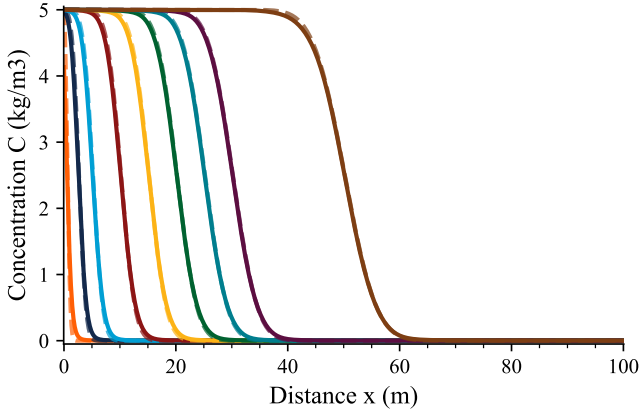
The next stage of the work is therefore methodological rather than purely accuracy-driven. The main objective is to identify a training strategy that is both efficient and consistently effective across the three model classes. This will require a more systematic study of collocation design, resampling frequency, optimizer scheduling, and overall model simplification, with the goal of moving from strong empirical results to a more justified and reproducible training protocol.

References

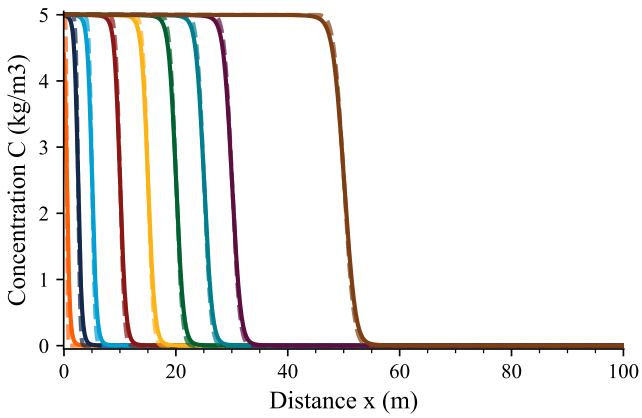
- [1] Akio Ogata and Robert B. Banks. *A Solution of the Differential Equation of Longitudinal Dispersion in Porous Media*. Professional Paper 411-A. U.S. Geological Survey, 1961. URL: <https://pubs.usgs.gov/pp/0411a/report.pdf> (visited on 01/01/2025).
- [2] Lu Lu et al. “Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators”. In: *Nature Machine Intelligence* 3.3 (Mar. 2021), pp. 218–229. ISSN: 2522-5839. DOI: [10.1038/s42256-021-00302-5](https://doi.org/10.1038/s42256-021-00302-5). URL: <http://dx.doi.org/10.1038/s42256-021-00302-5>.



(a) $Pe = 50$

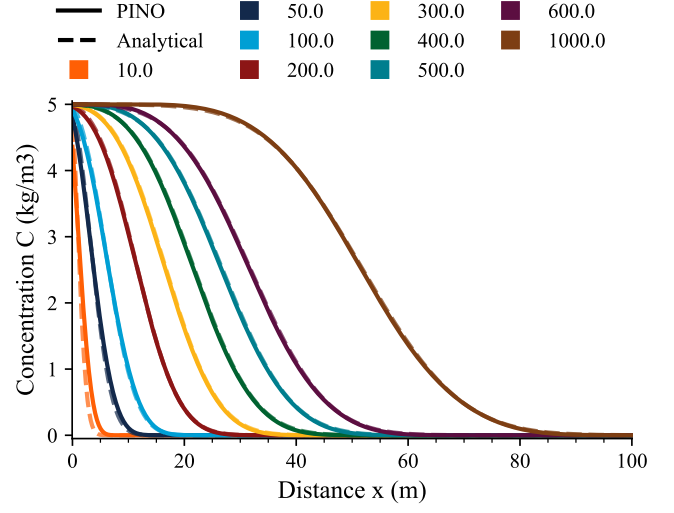


(b) $Pe = 500$

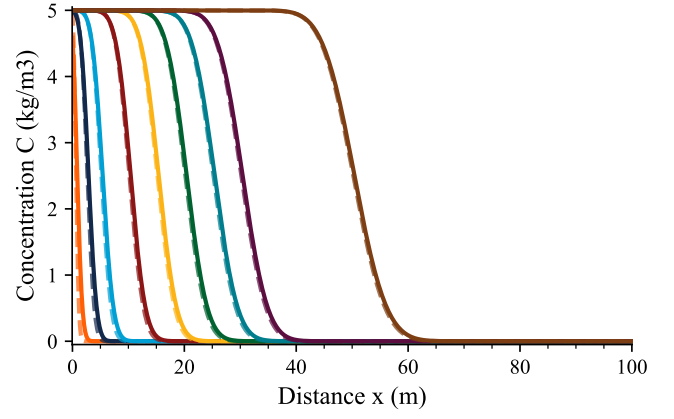


(c) $Pe = 5000$

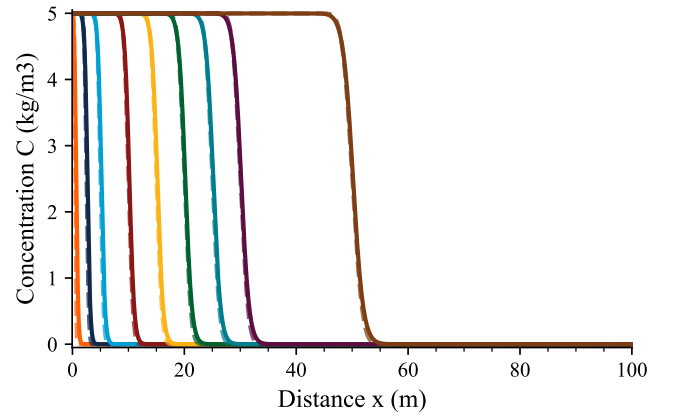
Figure 2: Parametric PINN: each subfigure shows results at a single Péclet number ($Pe = 50, 500$, and 5000).



(a) $Pe = 50$



(b) $Pe = 500$



(c) $Pe = 5000$

Figure 3: Parametric PINO (DeepONet): same layout as Figure 2—one Péclet number per subfigure ($Pe = 50, 500$, and 5000).